LITERATURE CITED

1. A. A. Bondarev, I. K. Dibobes, and K. I. Pyuskyulyan "Evaluation of the radiation conditions in the region of an atomic power plant under uncontrolled emission of radionuclides into the environment," At. Énerg., 60, No. 2, 138 (1986).
2. I. N. Ruzhentsova, R. V. Semova, V. N. Teverovskii, et al., "Characteristic features of the spread, in the surface air layer, of radionuclides ejected during regeneration of fuel elements," At. Énerg., 59, No. 5, 373 (1985).
3. K. Keller, Radiochemistry [in Russian], Atomizdat, Moscow (1978).
4. A. S. Belitskii and E. I. Orlova, Protection of Underground Water Against Radioactive Contaminations [in Russian], Meditsina, Moscow (1968).
5. V. M. Gol'dberg and S. Gazda, Hydrogeological Foundations of the Protection of Underground Water Against Contamination [in Russian], Nedra, Moscow (1984).
6. E. A. Yakovlev and A. I. Ostavnenko, "Methodology of hydrogeochemical monitoring of the geological environment in regions around atomic power plants," Razvedka Okhr. Nedr., No. 2, 35 (1988).

CHARACTERISTICS OF 96 UNDERGROUND NUCLEAR EXPLOSIONS AT THE SEMIPALATINSK PROVING GROUND

V. S. Bocharov, S. A. Zelentsov, and V. N. Mikhailov

UDC 550.834

Soviet seismologists consider the teleseismic method as an effective means of verifying compliance with treaties on the limitation of underground nuclear testing. It has been shown in [1, 2] that this method ensures high accuracy in estimating the explosion yield at proving grounds with known geophysical data concerning the ground at the test location, a sufficient volume of statistical observation data, and a permanent monitoring network. The experimental estimates of errors in determining the explosion yield at domestic proving grounds amount to less than 30% for a 0.95 confidence interval.

An increasing number of American scientists have begun to share lately the view that the teleseismic method is a highly efficient means of verification. They also maintain that, by using data on nuclear explosions with known yields, it is possible to establish a relationship between the seismic magnitude and the explosion yield. This relationship can be used for estimating unknown explosion yields. The indeterminancy connected with estimates of the systematic error can be reduced by reaching an agreement on limiting the test locations to special proving grounds and calibrating these grounds. If such calibration is made part of any future treaty, concern regarding systematic errors would be reduced to a minimum. Consequently, calibration of proving grounds is the foremost and most important part of the measures to be taken for reducing the indeterminacy in estimating the explosion yield [3].

The joint Soviet–American experiment on verification (SEV), performed at the Nevada and the Semipalatinsk proving grounds in 1988, provided a convincing example of calibration of proving grounds in the USSR and the USA. In the course of SEV, both sides obtained sufficient information on the efficiency of the monitoring methods used, in particular, the seismic method, and their applicability in practice.

One obviously cannot claim that any single, even carefully planned, experiment involving the highest methodic standards cannot provide statistically significant results. Realizing this, the USSR and the USA exchanged, during the SEV preparation, data on the yields and geophysical conditions relating to five other underground nuclear explosions in Nevada and at Semipalatinsk.

However, one must take into account the dissatisfaction of American scientists in connection with the fact that, except for the above-mentioned give explosions and the SEV explosion at the Semipalatinsk proving ground, the Soviet Union has not published data on its

Translated from Atomnaya Énergiya, Vol. 67, No. 3, pp. 210-214, September, 1989. Original article submitted February 25, 1989.

0038-531X/89/6703-0691$12.50 © 1990 Plenum Publishing Corporation

TABLE 1. Characteristics of Underground Nuclear Explosions at the Semipalatinsk Proving Ground

Date (Greenwich)	Time (Greenwich), hrs, minutes, seconds	Yield, Kt	Depth of loc. (line least resist.) of nuc. device, m	Geo. coor. (Lat. N, Long. E) of explosion epicenter, °, min, sec	Predominant surrounding rocks at the explosion site	Area	Remark
11.10.61	7 39 59,9	up to 20	116	49 46 21,8 77 59 42,0	Granite	Degelen	Tunnel
02.02.62	8 00 00,2	up to 20	238	49 46 38,9 78 00 05,9	»	»	»
15.03.64	8 00 00,4	20—150	220	49 48 57,5 78 04 30,6	»	»	»
16.05.64	6 00 59,8	20—150	253	49 48 27,8 78 06 07,1	»	»	»
19.07.64	6 00 00,6	up to 20	168	49 48 32,7 78 05 34,5	»	»	»
16.11.64	6 00 00,2	20—150	194	49 48 31,4 78 08 00,4	Quartz porphyrite	»	»
15.01.65	6 00 00,8	100—150	178	49 56 06,0 79 00 33,7	Sandstone	Balapan	Well with ejection [4]
03.03.65	6 14 59,4	up to 20	196	49 49 29,0 78 03 09,6	Granite	Degelen	Tunnel
11.05.65	6 40 00,2	up to 20	103	49 46 12,8 77 59 39,4	»	»	»
17.06.65	3 45 00,0	up to 20	152	49 49 42,1 78 04 00,7	»	»	»
29.07.65	3 05 00,2	up to 20	126	49 46 47,0 77 59 53,1	»	»	»
17.09.65	4 00 00,1	up to 20	156	49 48 41,7 78 08 48,1	Quartz porphyry	»	»
08.10.65	6 00 00,4	up to 20	204	49 49 33,3 78 06 41,2	Same	»	»
14.10.65	4 00 00,2	1,1	48	49 59 26,3 77 38 08,6	Siltstone	Murzhik	Well with ejection [4]
21.11.65	4 58 00,0	29	278	49 49 09,1 78 03 48,9	Granite	Degelen	Tunnel
24.12.65	5 00 00,2	up to 20	213	49 48 16,2 78 06 24,0	Quartz porphyry Same	»	»
13.02.66	4 58 00,1	125	297	49 48 32,2 78 07 15,6		»	»
20.03.66	5 50 00,3	100	294	49 45 41,9 78 01 26,0	» »	»	»
21.04.66	3 58 00,1	up to 20	178	49 48 34,8 78 06 00,1	Granite	»	»
07.05.66	3 58 00,2	4	274	49 44 34,3 78 06 17,9	Quartz porphyry	»	»
29.06.66	6 58 00,5	20—150	187	49 50 03,9 78 04 24,1	Granite	»	»
21.07.66	3 58 00,0	up to 20	176	49 44 12,0 78 05 49,3	Quartz porphyry	»	»
05.08.66	3 57 59,6	up to 20	171	49 45 51,5 78 02 32,7	Granite	»	»
19.08.66	3 52 59,9	up to 20	134	49 49 37,5 78 06 31,5	Quartz porphyry	Degelen	Tunnel
07.09.66	3 51 59,7	up to 20	117	49 49 43,8 78 03 49,5	Granite	»	»
19.10.66	3 57 59,9	20—150	185	49 44 49,6 78 01 13,9	»	»	»
03.12.66	5 02 00,2	up to 20	153	49 44 48,8 78 02 00,1	»	»	»
18.12.66	4 58 00,0	20—150	427	49 55 28,5 77 44 50,0	Porphyrite	Murzhik	Well
30.01.67	4 01 59,5	up to 20	131	49 46 02,8 77 59 29,0	Quartz syenite	Degelen	Tunnel
26.02.67	3 57 59,8	20—150	241	49 44 44,5 78 04 56,3	Quartz porphyry	»	»
25.03.67	5 58 01,1	up to 20	152	49 45 13,0 78 03 46,8	Granite	»	»
20.04.67	4 08 01,0	20—150	225	49 44 29,8 78 06 19,5	Quartz porphyry	»	»
28.05.67	4 07 59,6	up to 20	262	49 45 23,1 78 01 00,8	Same	»	»

nuclear tests (besides the peaceful nuclear explosions [4]), which reduced the statistical representability of experimental seismic data and caused doubts regarding estimates of the accuracy of the verification methods used.

TABLE 1 (continued)

Date (Greenwich)	Time (Greenwich), hrs, minutes, seconds	Yield, Kt	Depth of loc. (line least resist.) of nuc. device, m	Geo. coor. (Lat. N, Long. E) of explosion epicenter, °, min, sec	Predominant surrounding rocks at the explosion site	Area	Remark
29.06.67	2 56 59,9	up to 20	195	49 49 00,1 78 02 56,5	Granite	Degelen	Tunnel
15.07.66	3 26 59,9	up to 20	161	49 50 09,3 78 07 05,4	Quartz porphyry	»	»
04.08.67	6 58 00,3	up to 20	160	49 45 37,0 78 03 19,8	Granite	»	»
16.09.67	4 04 00,3	up to 20	230	49 56 13,9 77 43 41,2	Sandstone	Murzhik	Well
22.09.67	5 04 00,0	10	229	49 57 34,7 77 41 27,8	Siltstone	»	»
17.10.67	5 04 00,2	20—150	181	49 46 51,2 78 00 13,8	Granite	Degelen	Tunnel
30.10.67	6 04 00,0	up to 20	173	49 47 39,7 78 00 28,3	»	»	»
22.11.67	4 03 59,9	up to 20	227	49 56 31,0 77 41 12,6	Siltstone	Murzhik	Well
08.12.67	6 03 59,8	up to 20	150	49 49 01,7 78 09 49,6	Quartz porphyry	Degelen	Tunnel
07.01.68	3 46 59,9	up to 20	237	49 45 15,9 78 01 51,4	Granite	»	»
24.04.68	10 35 59,7	up to 20	127	49 50 42,7 78 06 11,6	Quartz porphyry	»	»
11.06.68	3 05 59,7	up to 20	149	49 47 34,8 78 08 42,3	Same	»	»
19.06.68	5 05 59,8	up to 20	316	49 58 48,9 78 59 07,8	Sandstone	Balapan	Well
12.07.68	12 08 00,0	up to 20	172	49 45 16,9 78 05 23,8	Granite	Degelen	Tunnel
20.08.68	04 05 59,6	up to 20	208	49 49 21,5 78 04 28,1	»	»	»
05.09.68	04 05 59,6	up to 20	162	49 44 29,8 78 04 32,1	»	»	»
29.09.68	03 43 00,0	60	290	49 48 43,1 78 07 19,0	Quartz porphyry	»	»
21.10.68	03 52 00,0	0,2	31,4	49 43 40,3 78 29 10,6	Argillite	Murzhik	Well with ejection [4]
09.11.68	2 54 00,1	up to 20	125	49 48 01,9 78 08 20,8	Quartz porphyry	Degelen	Tunnel
12.11.68	7 30 00,0	0,2×3	31,4	49 42 44,8 78 27 40,8	Gritstone	Murzhik	Grouped nuclear explosion of three charges with ejection[4]
18.12.68	5 01 59,7	up to 20	194	49 44 45,4 78 05 31,3	Granite	Degelen	Tunnel
07.03.69	8 26 59,8	20—150	214	49 49 17,3 78 03 45,6	»	»	»
16.05.69	4 02 59,7	up to 20	184	49 45 33,9 78 04 32,8	»	»	»
31.05.69	5 01 59,4	up to 20	258	49 57 01,1 77 41 39,2	Siltstone	Murzhik	Well
04.07.69	2 46 59,6	up to 20	219	49 44 45,7 78 06 40,8	Quartz porphyry	Degelen	Tunnel
23.07.69	2 47 00,2	16	175	49 48 56,3 78 07 46,6	Same	»	»
11.09.69	4 02 00,0	up to 20	190	49 46 34,7 77 59 48,1	Granite	»	»
01.10.69	4 02 59,9	up to 20	144	49 46 57,0 78 05 53,9	»	»	»
30.11.69	3 32 59,7	125	472	49 55 27,4 78 57 20,7	Conglomerate	Balapan	Well
28.12.69	3 47 00,2	46	388	49 56 14,4 77 42 51,2	Siltstone	Murzhik	»
29.12.69	4 02 00,0	up to 20	86	49 44 01,2 76 06 08,1	Quartz porphyry	Degelen	Tunnel
29.01.70	7 03 00,0	20—150	214	49 47 44,1 78 07 26,0	Porphyrite	»	»

TABLE 1 (continued)

Date (Greenwich)	Time (Greenwich), hrs, minutes, seconds	Yield, kt	Depth of loc. (line least resist.) of nuc. device, m	Geo. coor. (Lat. N, Long. E) of explosion epicenter, °, min, sec	Perdominant surrounding rocks at the explosion site	Area	Remark
27.03.70	5 02 59,6	up to 20	138	49 44 52,1 77 59 56,3	Granite	Degelen	Tunnel
27.05.70	4 03 00,0	up to 20	66	49 43 52,7 78 05 55,0	Quartz porphyry	»	»
28.06.70	1 58 00,0	20—150	332	49 48 05,4 78 06 24,5	Granite	»	»
21.07.70	3 02 59,7	up to 20	225	49 57 08,7 77 40 22,4	Sandstone	Murzhik	Well
24.07.70	3 57 00,0	up to 20	154	49 48 35,0 78 07 42,2	Quartz porphyry	Degelen	Tunnel
06.09.70	4 02 59,9	up to 20	212	49 45 35,1 78 00 19,4	Granite	»	»
04.11.70	6 02 59,8	up to 20	249	49 59 21,2 77 45 44,8	Porphyrite	Murzhik	Well
17.12.70	7 01 00,0	up to 20	193	49 44 44,3 78 05 57,0	Granite	Degelen	Tunnel
22.03.71	4 33 00,3	20—150	283	49 47 54,5 78 06 32,3	»	»	»
25.04.71	3 32 59,9	90	296	49 46 06,7 78 02 02,1	»	»	»
25.05.71	4 03 00,4	up to 20	132	49 48 05,9 78 08 19,8	»	»	»
06.06.71	4 02 59,7	16	299	49 58 31,5 77 39 37,0	Siltstone	Murzhik	Well
19.06.71	4 04 00,1	up to 20	290	49 58 08,5 77 38 26,9	Porphyrite	»	»
30.06.71	3 56 59,8	up to 20	217	49 56 45,6 78 58 49,7	Conglomerate	Balapan	»
09.10.71	6 02 59,7	12	237	49 58 40,4 77 38 29,2	Siltstone	Murzhik	»
21.10.71	6 02 59,7	23	324	49 58 25,7 77 35 50,4	Sandstone	»	»
29.11.71	6 02 59,9	up to 20	203	49 44 36,3 78 04 42,6	Granite	Degelen	Tunnel
15.12.71	7 52 59,8	up to 20	115	49 49 35,0 77 59 50,3	»	»	»
30.12.71	6 21 00,2	20—150	249	49 45 36,1 78 02 13,7	»	»	»
10.02.72	5 03 00,0	16	295	50 01 27,4 78 52 41,1	Siltstone	Balapan	Well
10.03.72	4 56 59,8	up to 20	171	49 44 43,1 78 07 10,9	Quartz porphyry	Degelen	Tunnel
28.03.72	4 22 00,1	6	124	49 43 59,0 78 04 32,5	Same	»	»
07.06.72	1 28 00,0	20—150	208	49 49 36,3 78 06 55,7	» »	»	»
06.07.72	1 03 00,0	up to 20	81	49 44 15,0 78 06 36,2	» »	»	»
16.08.72	3 16 59,8	8	139	49 45 55,7 78 03 31,8	Granite	»	»
26.08.72	3 46 59,7	up to 20	285	49 58 55,1 77 42 59,8	Siltstone	Murzhik	Well
02.09.72	8 56 59,9	2	185	49 57 33,9 77 38 27,2	Sandstone	»	»
02.11.72	1 27 00,2	165	521	49 55 37,1 78 49 02,1	Siltstone	Balapan	»
10.12.72	4 27 00,0	20—150	264	49 49 09,8 78 03 29,6	Granite	Degelen	Tunnel
10.12.72	4 27 10,0	140	478	50 01 37,2 78 59 44,0	Tufaceous sandstone	Balapan	Well
28.12.72	4 27 00,0	up to 20	132	49 44 21,1 78 06 22,5	Quartz porphyry	Degelen	Tunnel

The present publication of data on a number of the underground nuclear explosions carried out in the USSR will improve statistical estimates of the accuracy of the seismic method.

Table 1 provides some of the most important characteristics of the 96 underground nuclear explosions carried out at the Semipalatinsk proving ground up to 1973 along with general information on the conditions under which the tests were conducted. These data have been taken from the previously unpublished official sources of the Ministry of Defense. The authors hope that this information will provide an important contribution to the statistical substantiation of estimates of the accuracy of seismic methods for determining the yield of underground nuclear explosions.

LITERATURE CITED

1. V. S. Bocharov, M. N. Georgievskii, V. V. Kirichenko, and A. B. Peshkov, "Method of estimating the yield of underground nuclear explosions with an allowance for their actual seismic effectiveness," At. Énerg., 65, No. 2, 109-114 (1988).
2. V. S. Bocharov, L. P. Vladimirskii, and A. M. Novikov, "Estimate of the accuracy in determining the yield of underground nuclear explosions according to the seismic method for known test conditions," At. Énerg., 65, No. 2, 114-119 (1988).
3. U.S. Congress, Office of Technology Assessment, Seismic Verification of Nuclear Testing Treaties, OTA-ISC-361, Washington, D.C., U.S. Government Printing Office, May, 1988.
4. Atomic Explosions for Peaceful Purposes, Collection of Articles [in Russian], Atomizdat, Moscow (1970).